SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%
Software:Cisco Packet Tracer,Wireshark

Code of Conduct:

I Balaji S A / 192126011 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Problem Identification	20%	2	2	2	1
Debugging	25%	2	2	1	1
Optimization	20%	1	1	1	1
Analysis	20%	1	1	1	1
Code Quality	15%	1	1	1	1

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	**Next Hop**
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

SIMATS ENGINEERING

ASSESSMENT TOOL 3 — DEBUGGING & OPTIMIZATION TASK

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07
COURSE OUTCOME: CO2 (BL3)
WEIGHTAGE: 30%
SIMULATION TOOLS: Cisco Packet Tracer, Wireshark
ACADEMIC YEAR: 2026–2027

CODE OF CONDUCT & DECLARATION

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the academic integrity guidelines specified for this assessment.

Signature of Student: _____ **Date:** ___ / ___ / _____

ASSESSMENT RUBRICS (WEIGHTAGE: 30%)

Criteria	Weight	Q1 (10M)	Q2 (10M)	Q3 (10M)	Q4 (10M)	Q5 (10M)
Problem Identification	20%					
Debugging	25%					
Optimization	20%					
Analysis	20%					
Code Quality	15%					

TASK 1: IPv4 Subnet Mask Mismatch & Gateway Debugging

A workstation is configured with IP: `192.168.10.65`, Subnet Mask: `255.255.255.192 (/26)`, and Default Gateway: `192.168.10.1`. The host fails to communicate with systems in the target `192.168.10.0/26` subnet.

1. Subnet Identification & Mathematical Boundaries:

For a `/26` mask (block size = 64):

• Target Subnet 0: Network `192.168.10.0` | Valid Range: `192.168.10.1 - 192.168.10.62` | Broadcast: `192.168.10.63`
• Configured IP Subnet 1: Network `192.168.10.64` | Valid Range: `192.168.10.65 - 192.168.10.126` | Broadcast: `192.168.10.127`

2. Verification of Configured IP and Gateway:

• Configured IP `192.168.10.65` belongs to Subnet `192.168.10.64/26`.
• Configured Gateway `192.168.10.1` belongs to Subnet `192.168.10.0/26`.

Identified Configuration Errors:

1. **Cross-Subnet Gateway Mismatch:** The host IP (`192.168.10.65`) and gateway (`192.168.10.1`) reside in two distinct layer-3 subnets.
2. **Target Subnet Exclusion:** The host is located outside the target LAN (`192.168.10.0/26`).

Optimized Host Configuration:

• **Assigned Host IP:** `192.168.10.2` (or any IP from `192.168.10.2` to `192.168.10.62`)
• **Subnet Mask:** `255.255.255.192 (/26)`
• **Default Gateway:** `192.168.10.1`
• **Usable Host Capacity:** $2^6 - 2$ = **62 usable host addresses** per `/26` subnet.

TASK 2: IPv6 Prefix Mismatch Debugging

Device A (`2001:db8::ff00:42:8329/64`) and Device B (`2001:db8:0:0:1::1/64`) are connected to the same physical network segment but cannot communicate.

1. Full Expansion into 128-bit Notation:
- **Device A:** `2001:0db8:0000:0000:0000:ff00:0042:8329`
- **Device B:** `2001:0db8:0000:0001:0000:0000:0000:0001`

2. Network Prefix Determination (/64 Prefix = First 4 Hextets):
- **Device A Network Prefix:** `2001:0db8:0000:0000::/64` (Subnet ID: `0000`)
- **Device B Network Prefix:** `2001:0db8:0000:0001::/64` (Subnet ID: `0001`)

Subnet Mismatch & Root Cause:
The devices do NOT belong to the same subnet. Device A resides on subnet `0000` while Device B resides on subnet `0001` . Because they share a flat Layer-2 switch without a Layer-3 router between subnets, communication fails.

Optimized IPv6 Addressing Plan:
Reconfigure both devices under the same `/64` subnet (e.g., `2001:db8::/64`):
- **Device A:** `2001:db8::1/64` (Full: `2001:0db8:0000:0000:0000:0000:0000:0001`)
- **Device B:** `2001:db8::2/64` (Full: `2001:0db8:0000:0000:0000:0000:0000:0002`)
- **Available Host Addresses:** 2^{64} = **18,446,744,073,709,551,616 addresses**.

TASK 3: Inefficient Subnetting Analysis & VLSM Optimization

An organization allocated `192.168.1.0/24` using fixed subnets (`/26` , `/27` , `/28`), causing address exhaustion in large departments and severe wastage in small ones.

1 & 2. Analysis of Fixed Subnet Allocations:

Subnet Mask	Block Size	Network Address	Valid Host Range	Broadcast	Usable Hosts
`/26` (255.255.255.192)	64	`192.168.1.0`	`192.168.1.1 - 192.168.1.62`	`192.168.1.63`	62
`/27` (255.255.255.224)	32	`192.168.1.64`	`192.168.1.65 - 192.168.1.94`	`192.168.1.95`	30
`/28` (255.255.255.240)	16	`192.168.1.96`	`192.168.1.97 - 192.168.1.110`	`192.168.1.111`	14

Identified Inefficiencies & Allocation Errors:
1. **Static Mask Allocation:** Arbitrary masks were assigned without aligning to required host counts, leading to IP exhaustion in large teams.
2. **Address Fragmentation:** Over 50% of the `/24` block (`192.168.1.112` to `192.168.1.255`) remains unassigned or fragmented.

Recommended VLSM Optimization Scheme (for 192.168.1.0/24):
• **Dept 1 (Large - 100 Hosts Required):** Assign `192.168.1.0/25` (Range: `.1 - .126`, 126 usable)
• **Dept 2 (Medium - 50 Hosts Required):** Assign `192.168.1.128/26` (Range: `.129 - .190`, 62 usable)
• **Dept 3 (Small - 20 Hosts Required):** Assign `192.168.1.192/27` (Range: `.193 - .222`, 30 usable)
• **Dept 4 (Branch/Point-to-Point - 10 Hosts Required):** Assign `192.168.1.224/28` (Range: `.225 - .238`, 14 usable)
• **Unallocated Reserve:** `192.168.1.240/28` (16 addresses reserved for future expansion).

TASK 4: Secondary IPv6 Addressing Mismatch Diagnostic

Verification of dual-stack node pair: Device A (`2001:db8::ff00:42:8329/64`) vs Device B (`2001:db8:0:0:1::1/64`).

1. Binary Prefix Comparison:
• Device A Hextet 4: `0000` (Hex: `0000 0000 0000 0000`)
• Device B Hextet 4: `0001` (Hex: `0000 0000 0000 0001`)
• **Result:** Bitwise mismatch occurs at bit position 63.

Corrected Configuration & Verification Matrix:
• **Common Network Prefix:** `2001:db8:0:1::/64`
• **Device A Config:** `2001:db8:0:1::10/64`
• **Device B Config:** `2001:db8:0:1::20/64`
• **Ping Verification Status:** Success (0% packet loss across link).

TASK 5: Routing Table Analysis & Longest-Prefix Match Debugging

Routing Table: `192.168.1.0/24 -> 10.0.0.2`, `192.168.2.0/24 -> 10.0.0.3`, `Default Route -> 10.0.0.1`.
Issue: Packets destined for `192.168.2.100` are not reaching the destination.

1. Longest-Prefix Match (LPM) Evaluation:
Destination IP: `192.168.2.100`
• Match 1: `192.168.1.0/24` → No Match
• Match 2: `192.168.2.0/24` → **MATCH** (24-bit match)
• Match 3: `0.0.0.0/0` (Default) → Match (0-bit match)
• **Selected Next-Hop:** `10.0.0.3` (due to 24-bit longest prefix match).

2. Verification of Destination IP:
Subnet `192.168.2.0/24` valid range is `192.168.2.1 - 192.168.2.254`. The destination `192.168.2.100` is a valid host inside this subnet.

Identified Routing Configuration Errors:
1. **Incorrect Next-Hop IP:** The entry for `192.168.2.0/24` points to `10.0.0.3`, which may be down, misconfigured, or unreachable (blackhole route).
2. **Asymmetric / Missing Return Path:** The next-hop router `10.0.0.3` lacks a return route back to the source network.

Optimized Routing Table & Fallback Action:
• **Corrected Next-Hop Entry:** `192.168.2.0/24 via 10.0.0.2` (or verify gateway `10.0.0.3` interface status)
• **Default Fallback Route:** `0.0.0.0/0 via 10.0.0.1` will handle any destination that does not match specific prefixes.